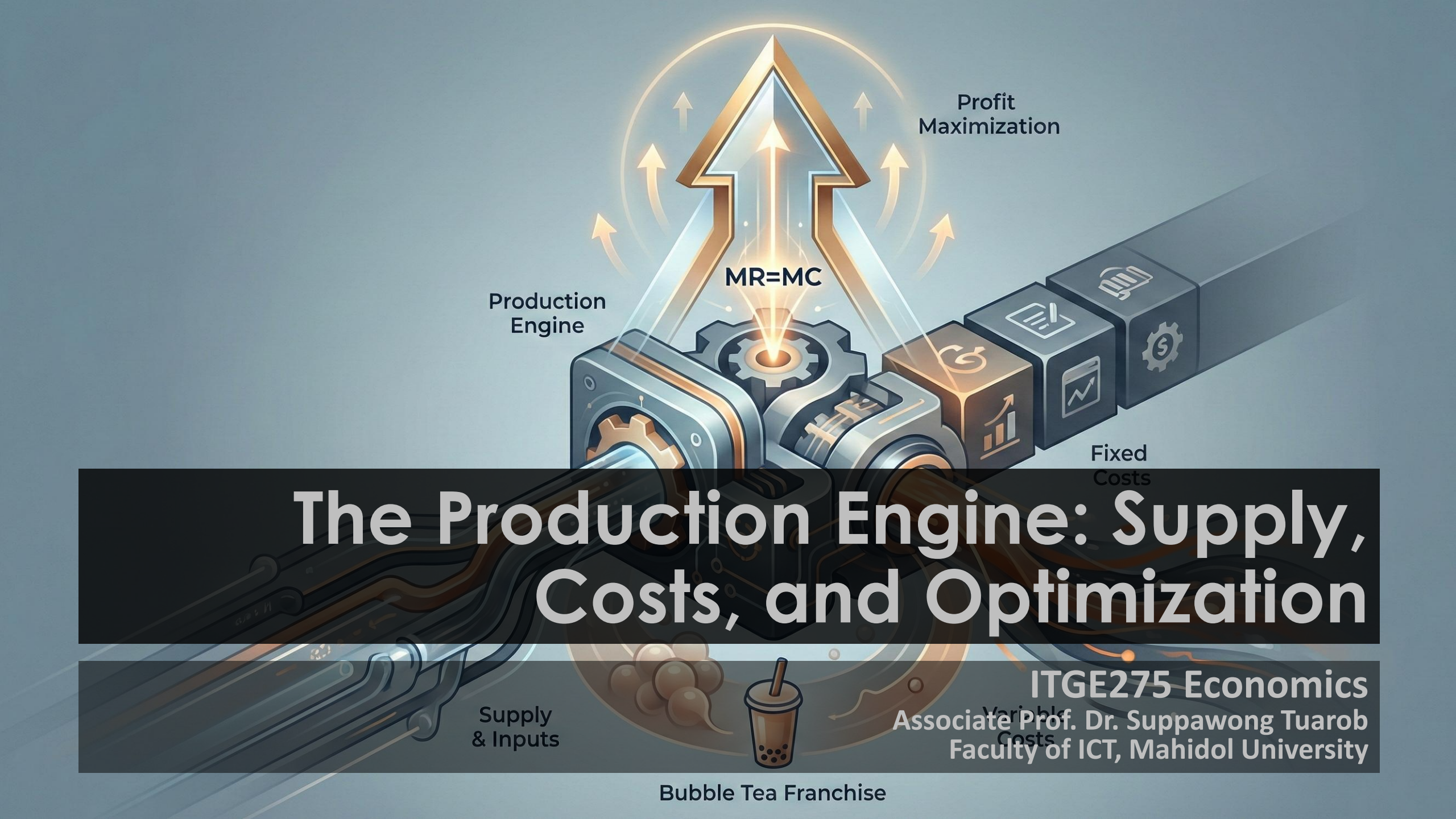


The Production Engine: Supply, Costs, and Optimization

ITGE275 Economics
Associate Prof. Dr. Suppawong Tuarob
Faculty of ICT, Mahidol University

The Economic Agent: The Firm

- **Concept:** In the previous session, we modeled the "Consumer" as an agent maximizing Utility. Today, we model the **"Firm"**.
- **Definition:** A Firm is an entity that converts inputs (resources) into outputs (goods/services).
- **The Black Box Model:** In CS terms, a firm is simply a function ***f(x)***.
  - **Input (*x*):** Factors of Production (Labor, Capital, Land).
  - **Process (*f*):** The Production Function (Technology).
  - **Output (*y*):** The Product.

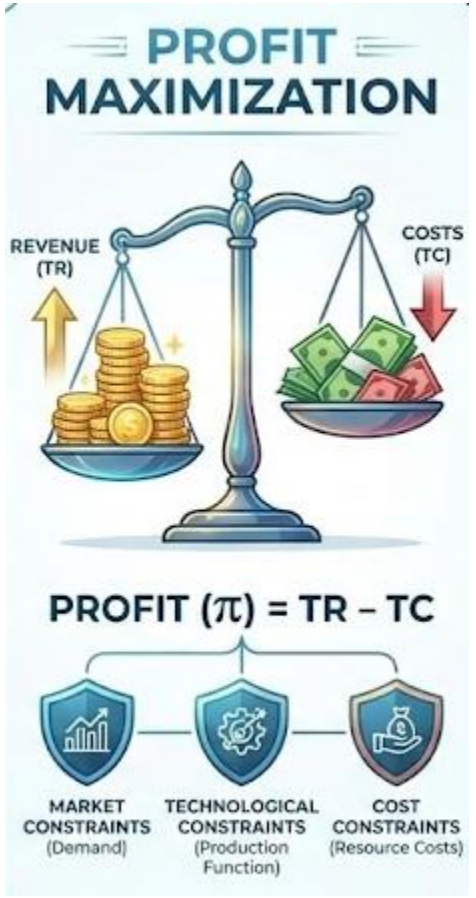


The Objective Function: Profit Maximization

- **The Algorithm:** Just as consumers optimize for Happiness, Firms optimize for **Profit (π)**.
- **The Equation:**

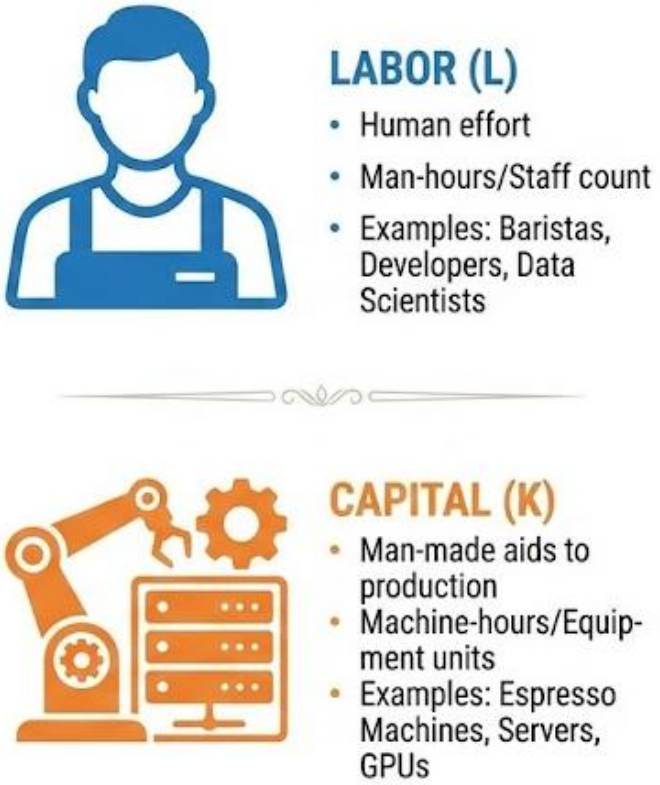
$$\pi = \text{Total Revenue (TR)} - \text{Total Cost (TC)}$$

- **Constraints:** The firm cannot produce infinite profit due to:
  - **Market Constraints:** Customers will only buy so much (Demand).
  - **Technological Constraints:** You can only squeeze so much output from your machines (Production Function).
  - **Cost Constraints:** Resources are not free.



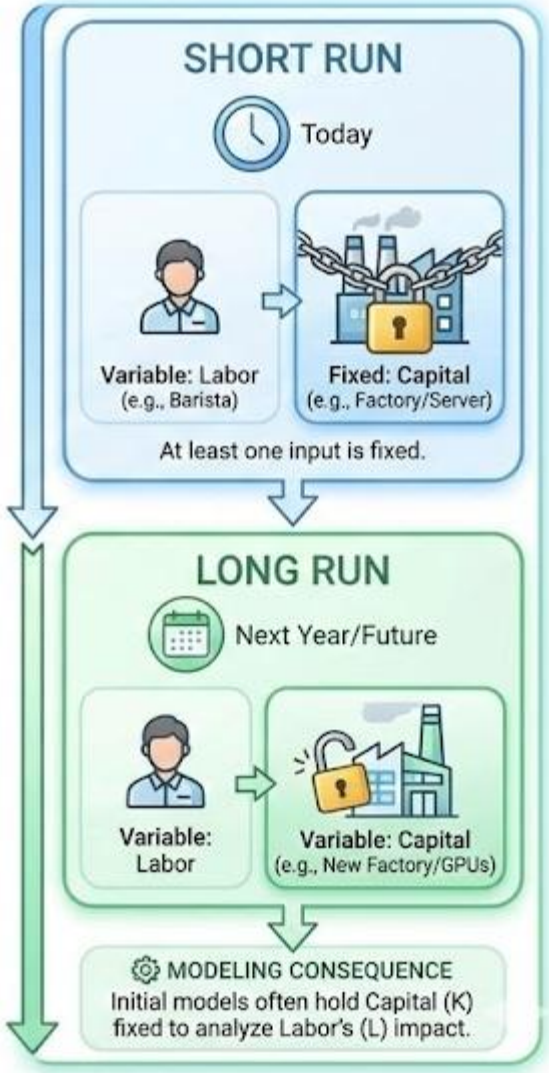
Factors of Production (Review)

- To build our model, we need variables. We group inputs into two main categories for simplicity:
  - **Labor (*L*):** Human effort. Measured in "Man-hours" or "Staff count."
    - *Example:* Baristas, Developers, Data Scientists.
  - **Capital (*K*):** Man-made aids to production. Measured in "Machine-hours" or units of equipment.
    - *Example:* Espresso Machines, Servers, GPUs.



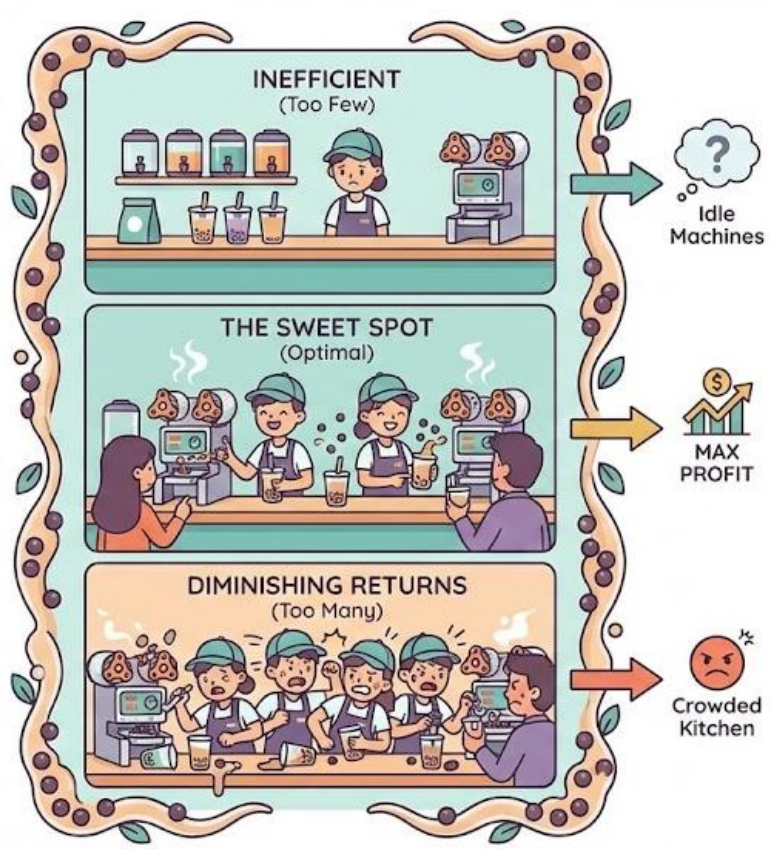
The Short Run vs. The Long Run

- **Short Run:** A period where at least one input is fixed.
  - *Example:* You can hire a new barista today (Variable ***L***), but you cannot build a new factory or buy a new server rack instantly (Fixed ***K***).
- **Long Run:** A period where *all* inputs are variable.
  - *Example:* You can move to a new office, buy 1000 GPUs, or restructure the entire company.
- **Modeling Consequence:** In our initial simulations, we often hold Capital (***K***) fixed to see the marginal impact of adding Labor (***L***).



Case Study: The Bubble Tea Franchise

- **Scenario:** A local entrepreneur wants to scale a Bubble Tea business.
- **The Problem:** How many staff members (L) and how many sealing machines (K) should be in each kiosk?
- **The Trade-off:**
  - Too few staff = Machines sit idle (Inefficient Capital).
  - Too many staff = Kitchen gets crowded, workers bump into each other (Diminishing Returns).
- **Goal:** Find the "Sweet Spot" using computational optimization.



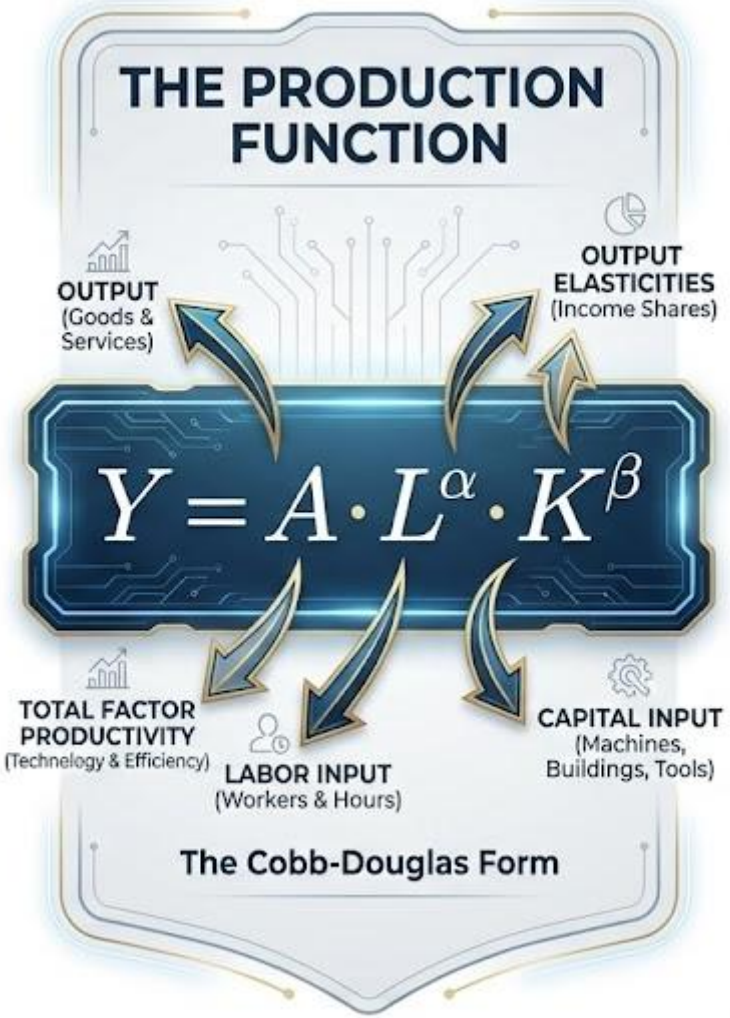
Introducing the Production Function

- **Mathematical Definition:** A relationship between the quantity of inputs and the maximum quantity of outputs.

$$Q = f(L, K)$$

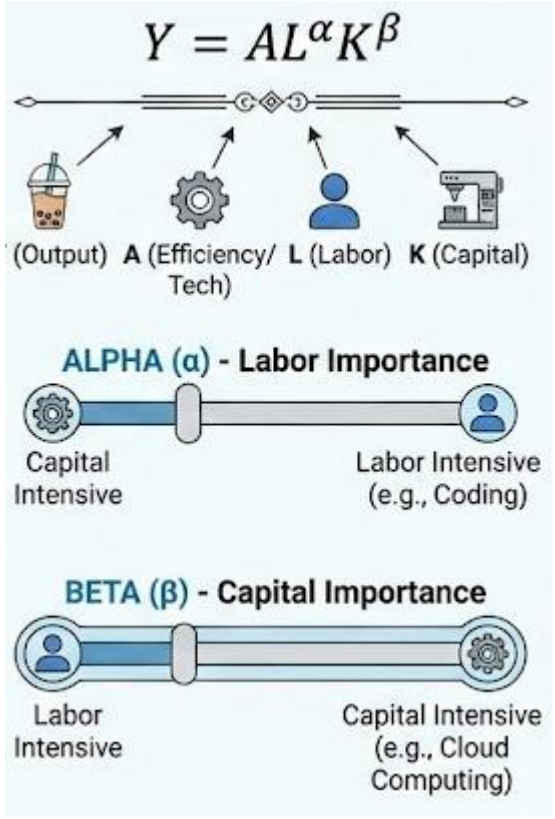
- **The Cobb-Douglas Function:** The most famous function in economics, widely used because it fits real-world manufacturing data surprisingly well.

$$Y = A \cdot L^{\alpha} \cdot K^{\beta}$$



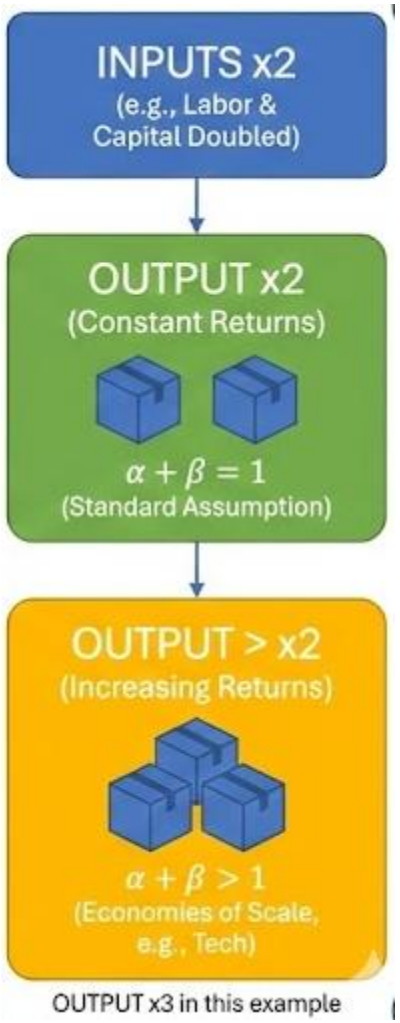
Deconstructing Cobb-Douglas ($Y = AL^{\alpha}K^{\beta}$)

- **Y (Output):** The number of bubble tea cups produced.
- **A (Total Factor Productivity):** Represents **Technology** or Efficiency.
  - If you invent a faster sealing machine, ***A*** increases. You produce *more* with the *same* inputs.
- **L & K:** Quantity of Labor and Capital.
- **α & β (Output Elasticities):**
  - These exponents determine how responsive output is to changes in inputs.
  - If α is high (e.g., 0.8), the industry is **Labor Intensive** (e.g., Massage Therapy, Coding).
  - If β is high (e.g., 0.8), the industry is **Capital Intensive** (e.g., Semiconductor Fab, Cloud Computing).



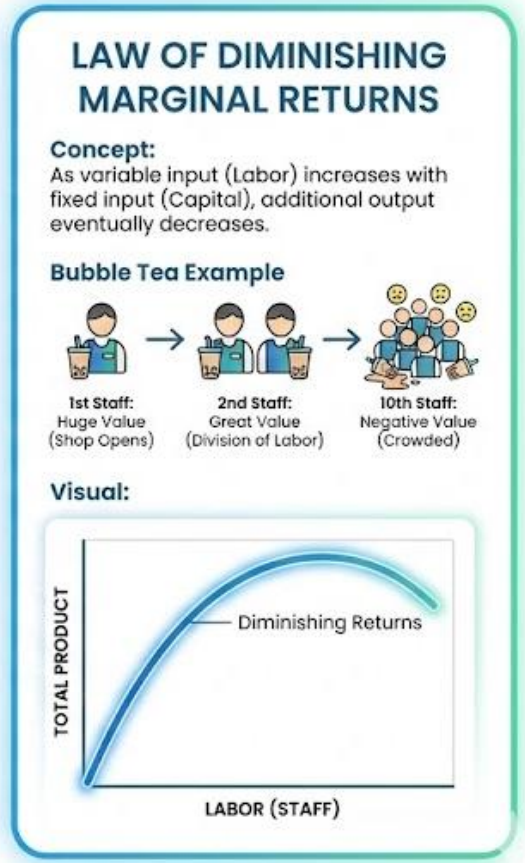
Returns to Scale

- What happens if we **double** both inputs?
- **Constant Returns to Scale (α + β = 1):**
  - Double inputs → Double output. (Standard assumption).
- **Increasing Returns to Scale (α + β > 1):**
  - Double inputs → *More* than double output. (Economies of Scale - common in Software/Tech).
- **Decreasing Returns to Scale (α + β < 1):**
  - Double inputs → *Less* than double output. (Bureaucracy/Inefficiency).



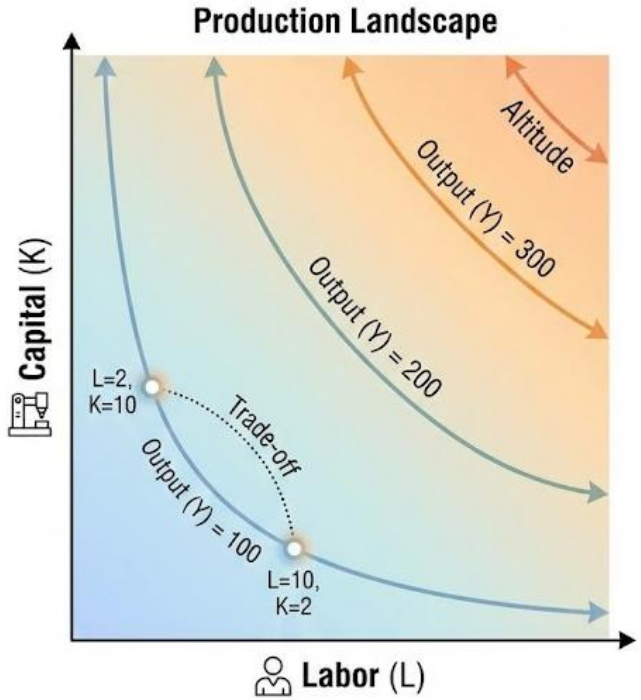
Law of Diminishing Marginal Returns

- **The Concept:** As you add more of a *variable* input (Labor) to a *fixed* input (Capital), the *additional* output eventually decreases.
- **Bubble Tea Example:**
  - 1st Staff: Huge value (Shop opens).
  - 2nd Staff: Great value (Division of labor).
  - ...
  - 10th Staff: Negative value (Kitchen is too crowded, spills tea).



Visualizing Production: Isoquants

- **Isoquant (Iso = Equal, Quant = Quantity):** A curve showing all combinations of ***L*** and ***K*** that produce the *same* amount of output.
- **Topographic Map Analogy:** Think of Isoquants as contour lines on a map. The "Altitude“ is the Output (***Y***).
- **Trade-off:** You can produce 100 cups with (10 Staff, 2 Machines) OR (2 Staff, 10 Machines).

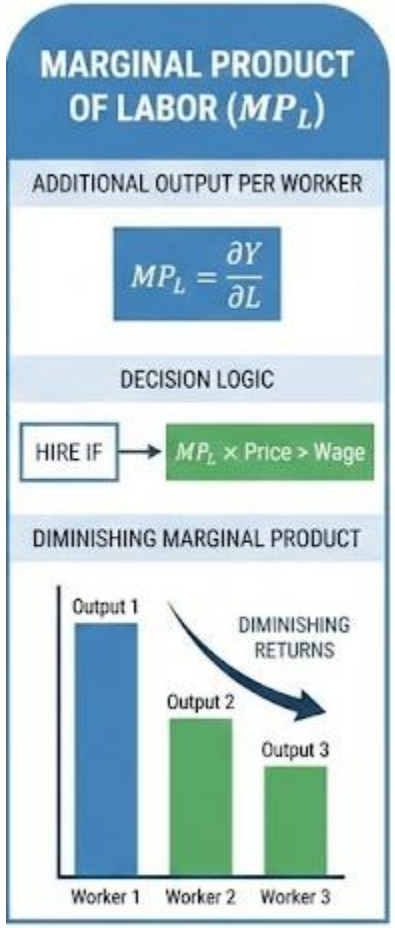


Marginal Product of Labor (MP_L)

- **Definition:** The additional output produced by hiring one more worker.
- **Calculus:** The partial derivative of the production function with respect to Labor.

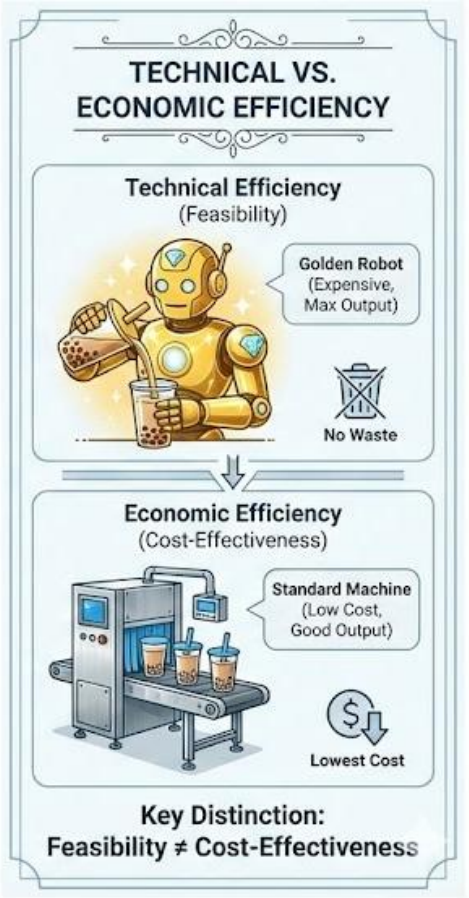
$$MP_L = \frac{\partial Y}{\partial L}$$

- **Decision Logic:** A firm only hires if the value of the worker's output (MP_L x Price) is greater than the worker's wage.



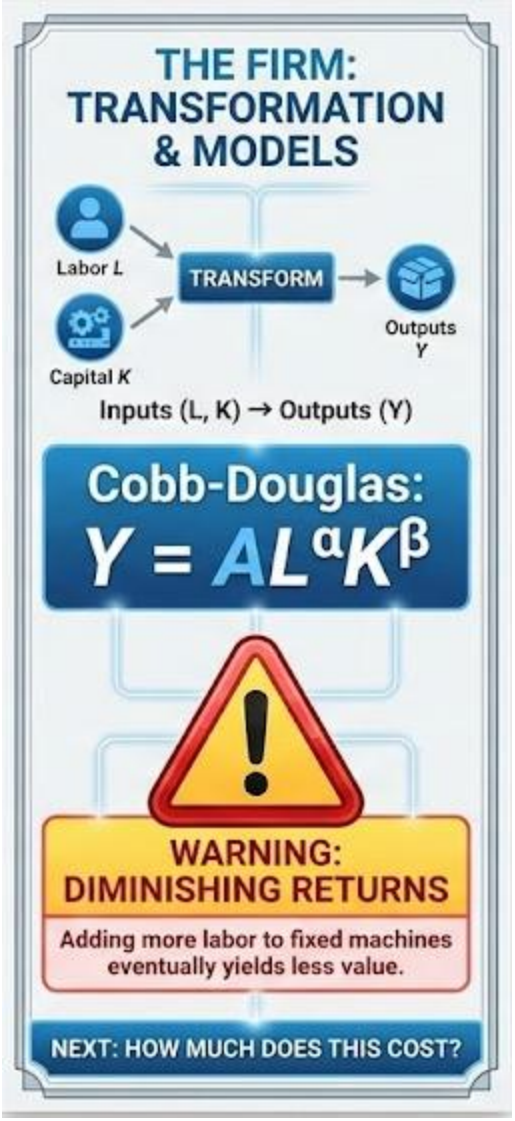
Technical Efficiency vs. Economic Efficiency

- **Technical Efficiency:** Producing the maximum output with given inputs (Not wasting tea).
- **Economic Efficiency:** Producing a given output at the **lowest possible cost**.
- **Key Distinction:** A process can be technically efficient (using a diamond-encrusted robot to make tea) but economically inefficient (too expensive).



Part 2 Summary

- Firms transform inputs (***L***, ***K***) into outputs (***Y***).
- Cobb-Douglas models this relationship: $Y = AL^{\alpha}K^{\beta}$
- **Diminishing Returns:** Adding more labor to fixed machines eventually yields less value.
- **Next:** How much does all this cost?



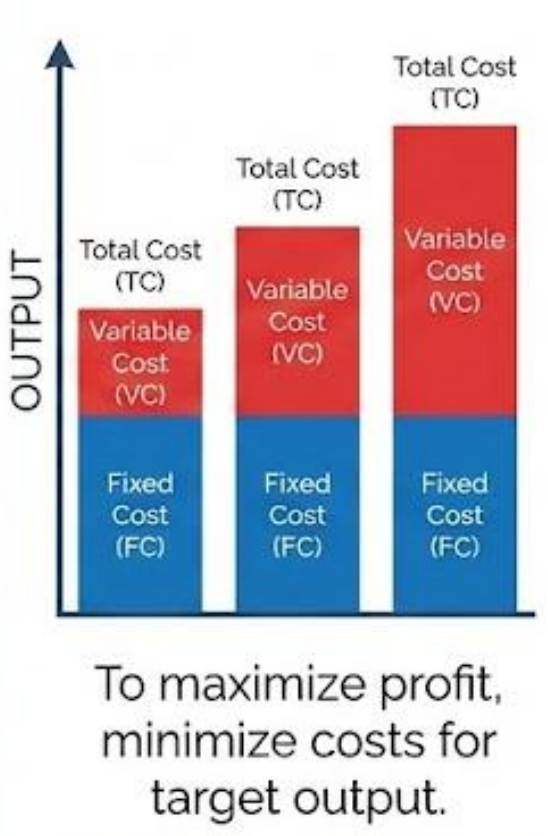
Part 3: Cost Analysis

The Cost Structure

- Production logic dictates Cost logic.
- **Total Cost (TC):** The sum of all costs incurred in production.

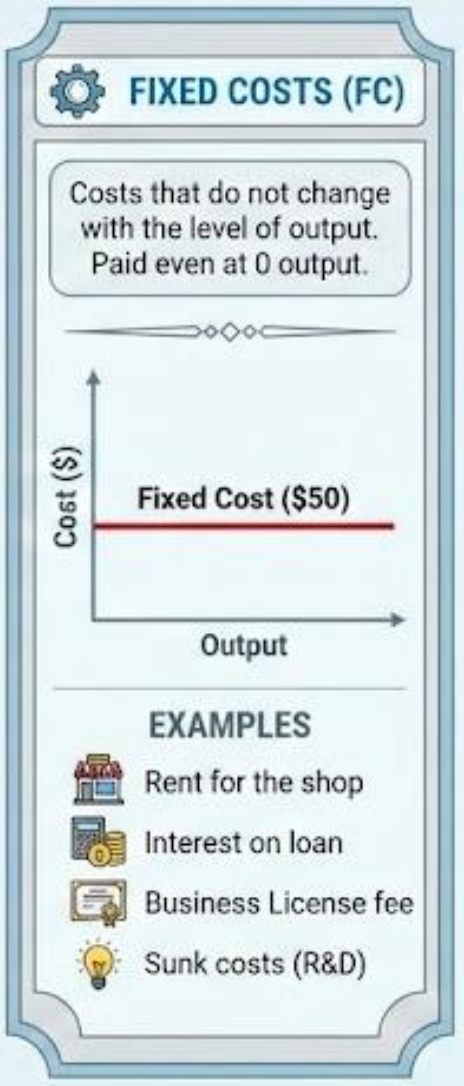
$$TC = \text{Fixed Costs (FC)} + \text{Variable Costs (VC)}$$

- **Relevance:** To maximize profit, we must minimize these costs for any given target output.



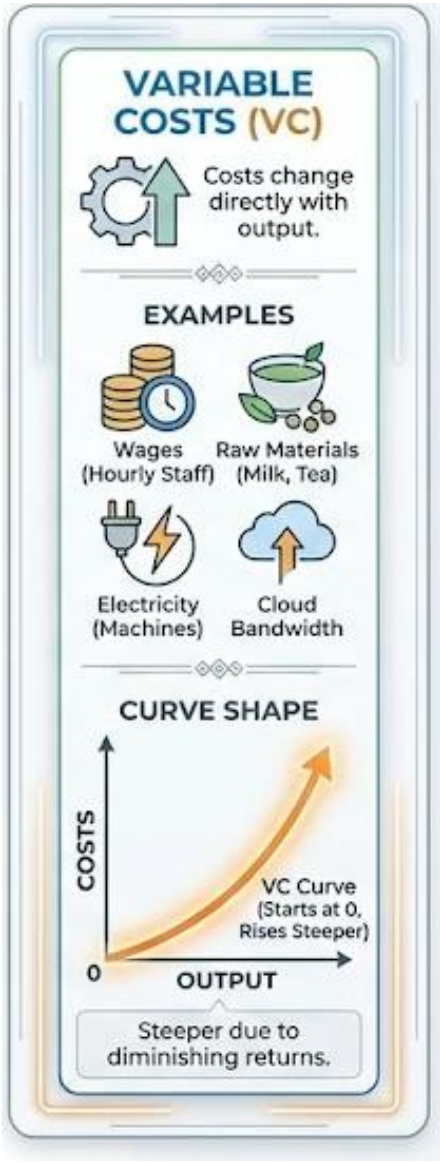
Fixed Costs (FC)

- **Definition:** Costs that **do not** change with the level of output. You pay these even if you produce 0 cups of tea.
- **Examples:**
  - Rent for the shop.
  - Interest on the loan for the espresso machine.
  - Annual Business License fee.
  - Sunk costs (Research & Development).
- **Curve Shape:** A flat horizontal line.



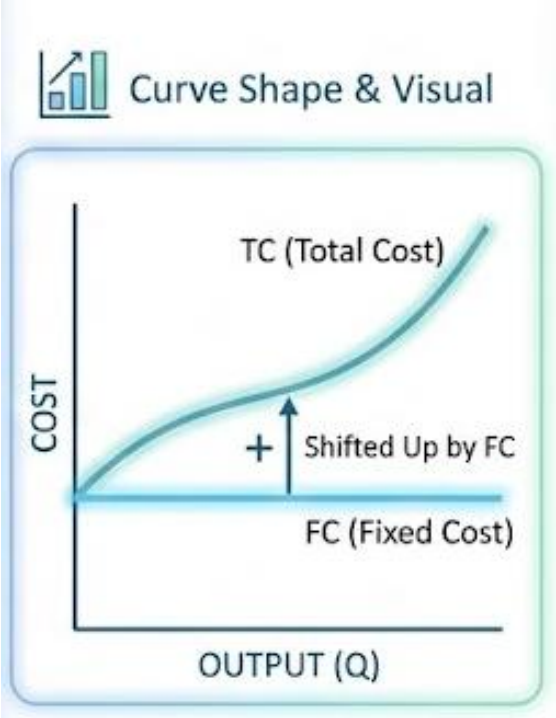
Variable Costs (VC)

- **Definition:** Costs that change directly with the level of output.
- **Examples:**
  - Wages for hourly staff.
  - Raw materials (Milk, Tea, Tapioca pearls).
  - Electricity (if machines run more).
  - Cloud Bandwidth (for tech firms).
- **Curve Shape:** Starts at 0 and rises. It gets steeper due to diminishing returns.

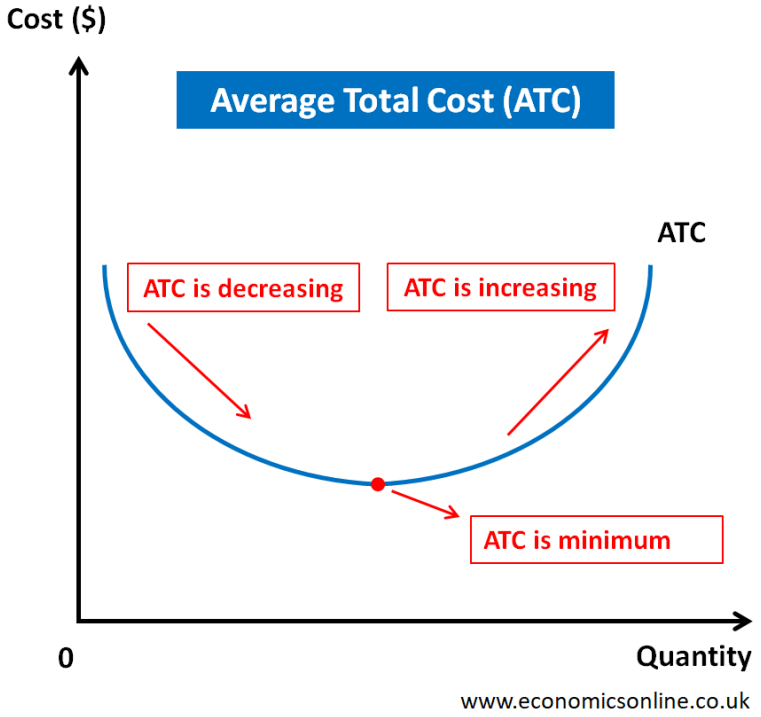


Total Cost (TC)

- **Definition:** The vertical sum of FC and VC.
- **Curve Shape:** Looks like the Variable Cost curve, but shifted up by the amount of the Fixed Cost.



Average Total Cost (ATC)



- **Definition:** The cost per unit of output.

$$ATC = \frac{TC}{Q}$$

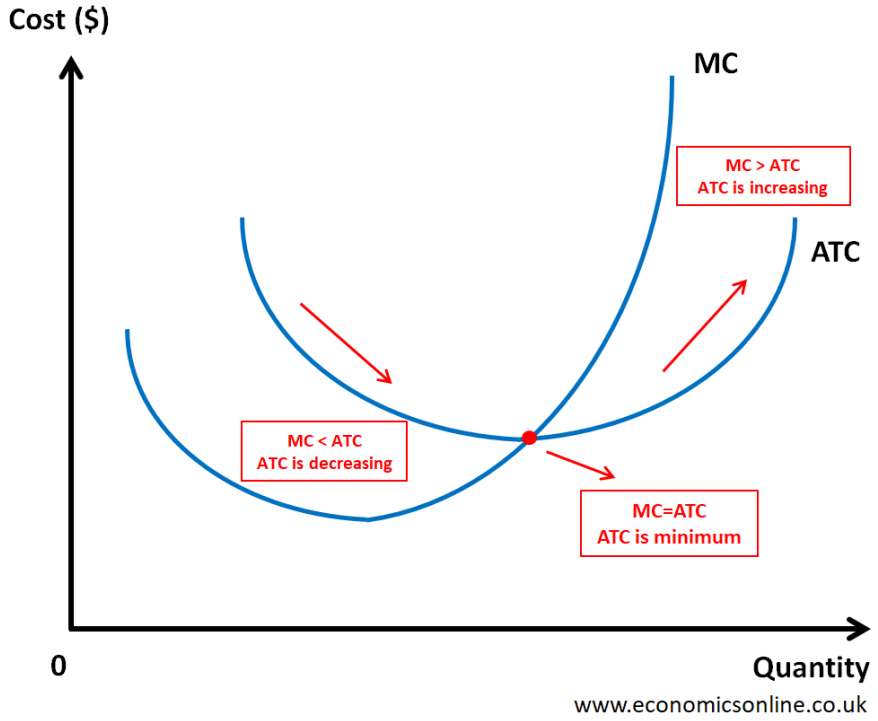
- **The "U" Shape:**
  - **Falling side:** Economies of scale (spreading fixed costs over more units).
  - **Rising side:** Diseconomies of scale (diminishing returns, overcrowding).
- **Business Goal:** Operate near the bottom of the "U" (Minimum Efficient Scale).

Marginal Cost (MC) - The Most Important Curve

- **Definition:** The cost of producing **one more** unit.

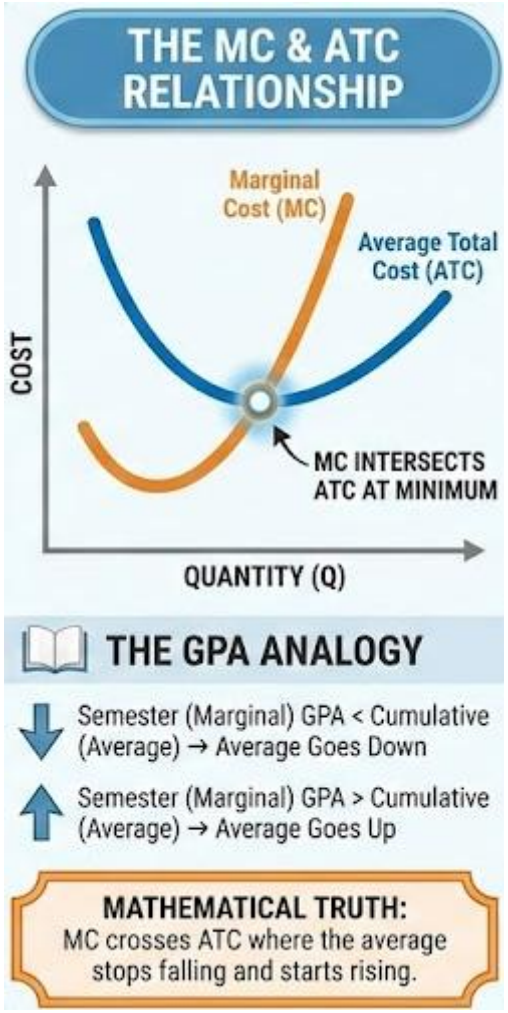
$$MC = \frac{\Delta TC}{\Delta Q}$$

- **Why is it vital?** Decisions are made at the margin. "Should I make cup #101?" depends on the cost of cup #101, not the average cost of the first 100.
- **Curve Shape:** "J" shaped. Falls briefly due to specialization, then rises sharply due to diminishing returns.



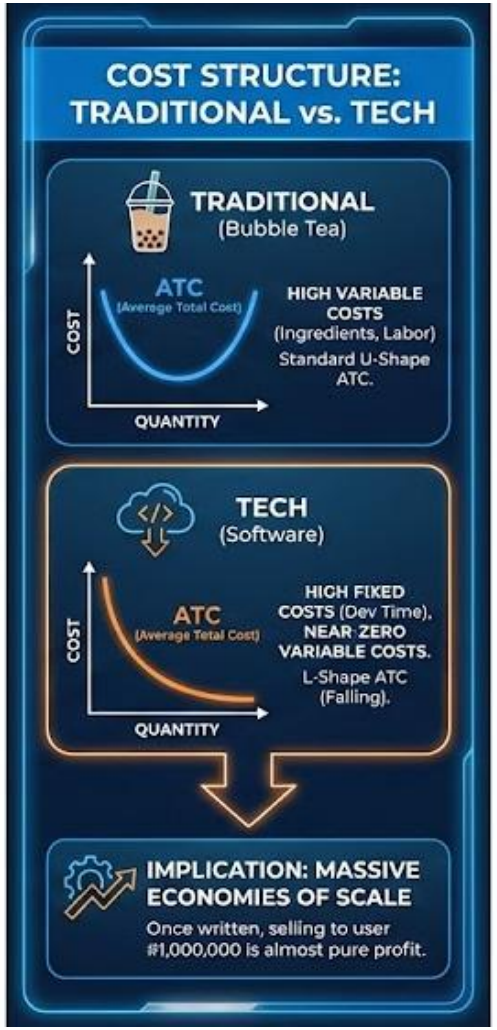
The Relationship between MC and ATC

- **Mathematical Truth:** The Marginal Cost curve always intersects the Average Total Cost curve at its **minimum point**.
- **Analogy:**
  - If your GPA for this semester (Marginal) is lower than your cumulative GPA (Average), your Average goes down.
  - If your GPA for this semester is higher than your cumulative GPA, your Average goes up.
  - Therefore, they meet exactly where the average stops falling and starts rising.



Cost Structure in Tech vs. Traditional

- **Traditional (Bubble Tea):** High Variable Costs (Ingredients, Labor).
- **Tech (E.g., Subscription Model):** High Fixed Costs (Dev Time), Near-Zero Variable Costs (Copying a file).
- **Implication:** Tech firms have massive Economies of Scale. Once the software is written, selling to user #1,000,000 is almost pure profit.



Re: Implicit vs. Explicit Costs (Opportunity Cost)

- **Explicit Costs:** Money actually leaving the bank (Wages, Rent).
- **Implicit Costs:** Opportunity costs of using resources you own (e.g., the owner's time, the forgone interest on capital).
- **Economic Profit:** Total Revenue - (Explicit + Implicit Costs).
- **Accounting Profit:** Total Revenue - Explicit Costs.
- **Homework Hint:** Remember this when analyzing the factory's choice between Product A and B.



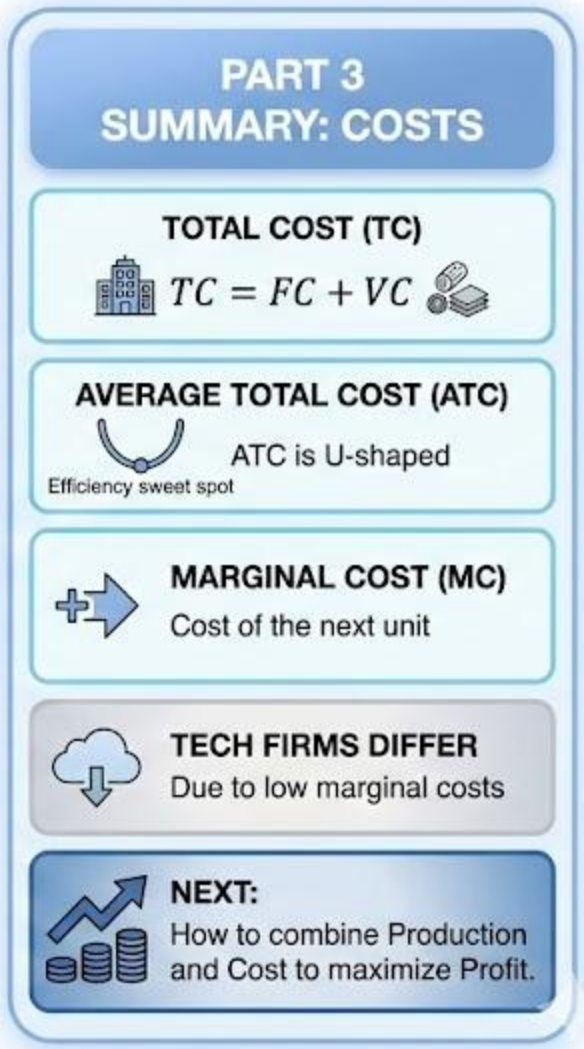
Re: Sunk Costs

- **Definition:** Costs that have already been incurred and cannot be recovered.
- **Rational Choice:** Sunk costs should be **ignored** in future decision-making.
- **Fallacy:** "We spent $1M on this failed server architecture, we must keep using it." (Wrong. The money is gone. Optimize for *now*).



Part 3 Summary

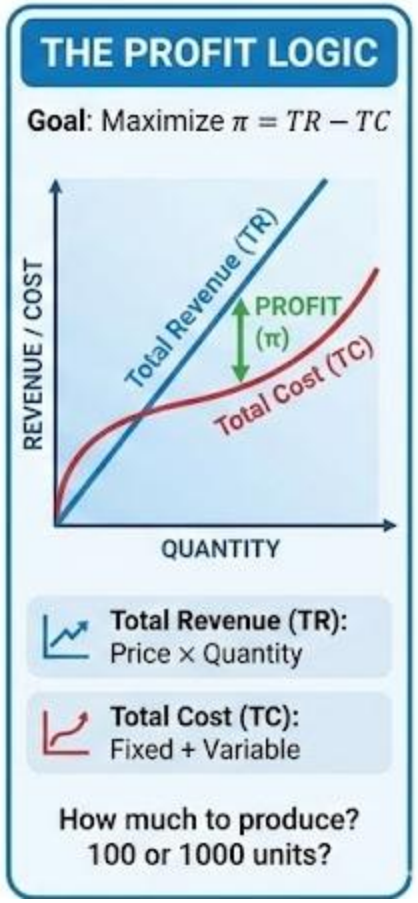
- TC = FC + VC
- ATC is U-shaped (Efficiency sweet spot).
- MC is the cost of the *next* unit.
- Tech firms differ due to low marginal costs.
- **Next:** How to combine Production and Cost to maximize Profit.



Part 4: Profit Maximization

The Profit Logic

- **Goal:** Maximize π = TR - TC.
- We know how to calculate TC (Cost Curves).
- **Total Revenue (TR):** Price × Quantity.
- How much should we produce? 100 units? 1000 units?



Marginal Revenue (MR)

- **Definition:** The additional revenue gained from selling **one more** unit.
- **In Competitive Markets:** MR = Price (***P***).
  - If you sell bubble tea for $5, the MR of the 100th cup is $5.



e.g., Bubble Tea Price = $5.
MR for 100th cup = $5.

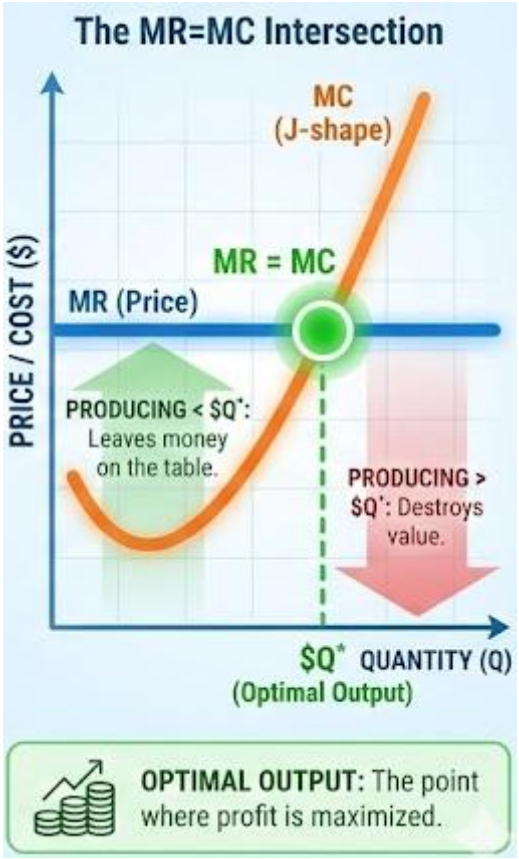
The Golden Rule of Business

- **The Rule:** Profit is maximized at the quantity where:
  - ***MR*** = ***MC***
- **Logic:**
  - If ***MR*** > ***MC***: You make profit on the next unit. **Keep producing.**
  - If ***MR*** < ***MC***: You lose money on the next unit. **Stop producing.**
  - Therefore, stop exactly where they are equal.



Visualizing the Optimal Quantity (Q^*)

- We plot the MR curve (Horizontal Price line) and the MC curve (J-shape).
- The intersection point determines the optimal quantity Q^*.
- Producing less than Q^* leaves money on the table.
- Producing more than Q^* destroys value.



The Shutdown Rule

- What if you are losing money (Profit < 0)?
- **Short Run Decision:**
  - If ***P*** < ***AVC*** (Price doesn't even cover variable costs like milk/cups): **Shut Down immediately.**
  - If ***P*** > ***AVC*** but ***P*** < ***ATC*** (You cover variable costs but not rent): **Keep operating** in the short run to minimize losses.
- **Long Run Decision:** Exit the market.



Economies of Scale

- **Definition:** Cost advantages reaped by companies when production becomes efficient.
- **Causes:**
  - Bulk buying discounts (Raw materials).
  - Specialization of Labor.
  - Spreading fixed costs (Software).
- **Impact:** Lowers the ATC curve, increasing the profit margin.









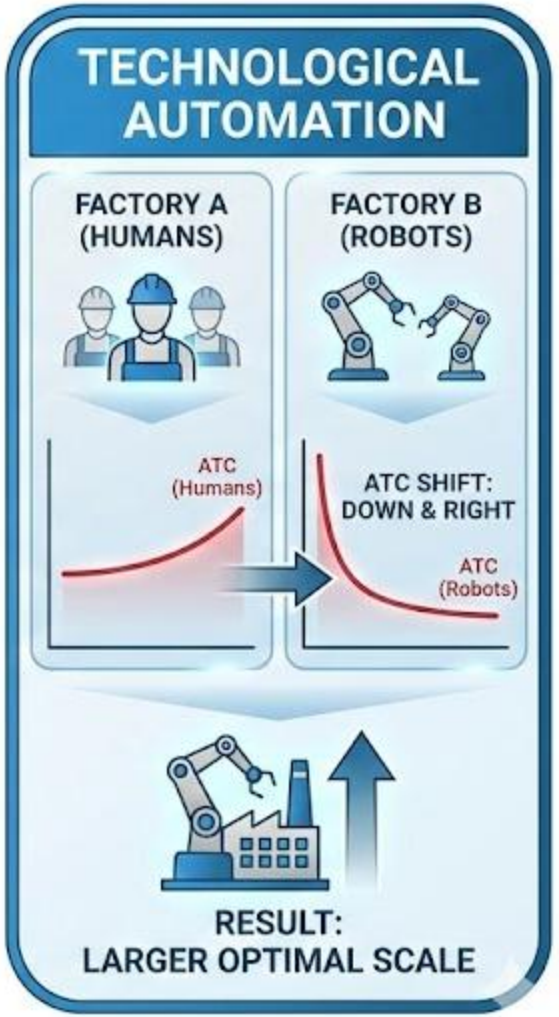






Technological Automation

- How does automation affect the curves?
- **Change:** Increases Fixed Costs (buying robots), but drastically decreases Variable Costs (no wages).
- **Result:** The ATC curve shifts down and to the right. The optimal scale becomes much larger.



Part 5: Computational Approach

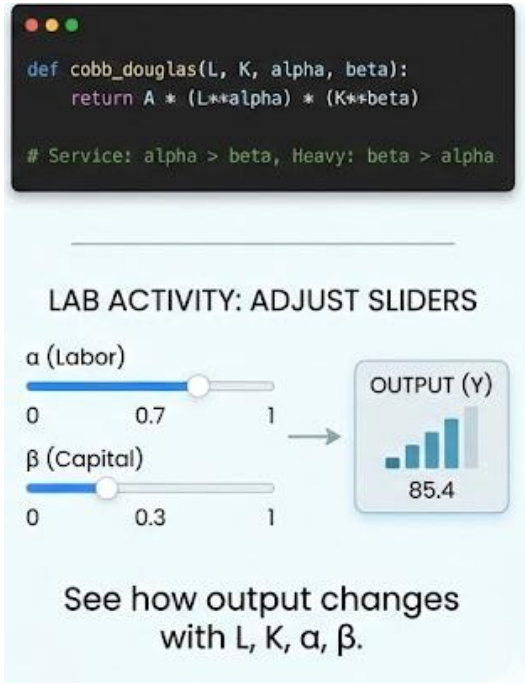
From Theory to Python

- In the Tutorial, we don't just draw curves; we **solve** for them.
- **Tools:** numpy for calculation, matplotlib for visualization, scipy.optimize for finding solutions.



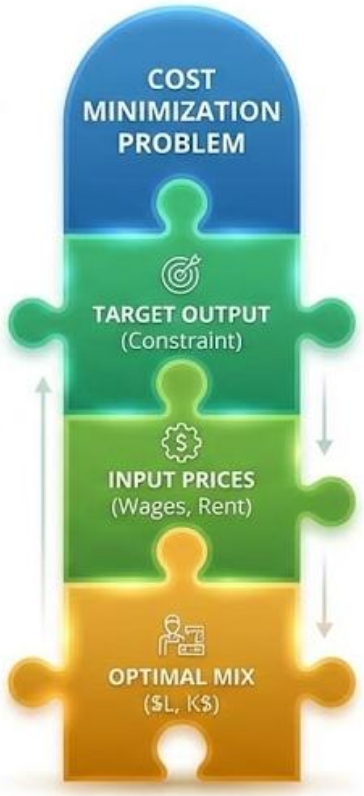
Simulating Cobb-Douglas

- We create a Python function cobb_douglas(L, K, alpha, beta).
- This allows us to simulate different industries:
  - Service Industry: Set α > β.
  - Heavy Industry: Set β > α.
- **Lab Activity:** You will adjust sliders to see how output changes.



Cost Minimization Problem

- **The Business Problem:** "I need to produce 100 cups of tea. What is the cheapest combination of Staff (***L***) and Machines (***K***)?"
- **Inputs:**
  - Target Output (Constraint).
  - Wage Rate (Cost of ***L***).
  - Machine Rent (Cost of ***K***).

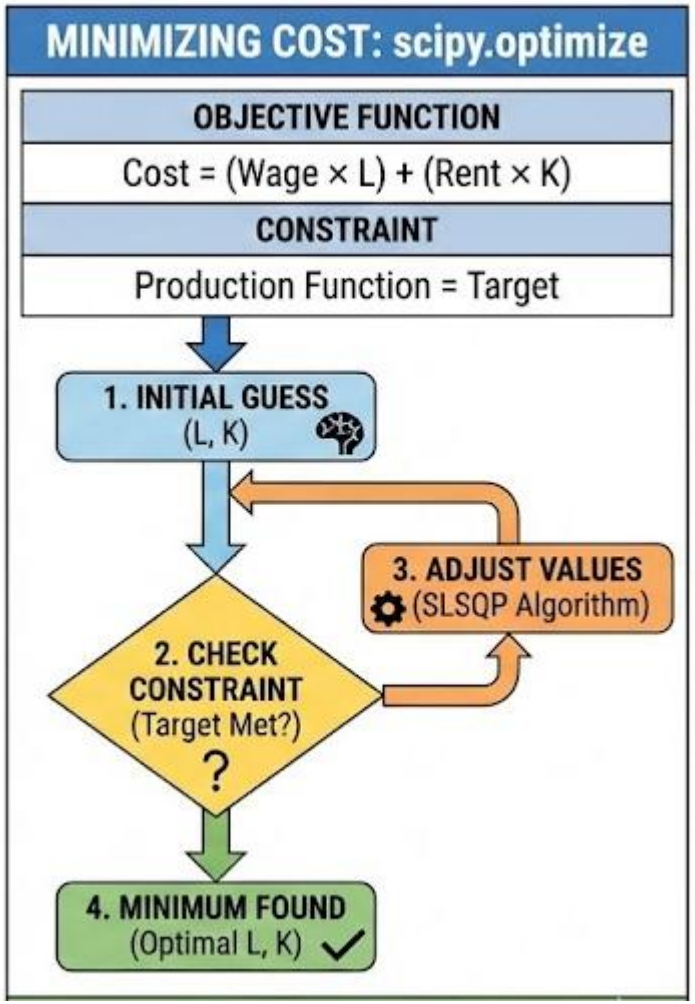


Using scipy.optimize.minimize

- We define an **Objective Function**: Cost = (Wage × L) + (Rent × K).
- We define a **Constraint**:

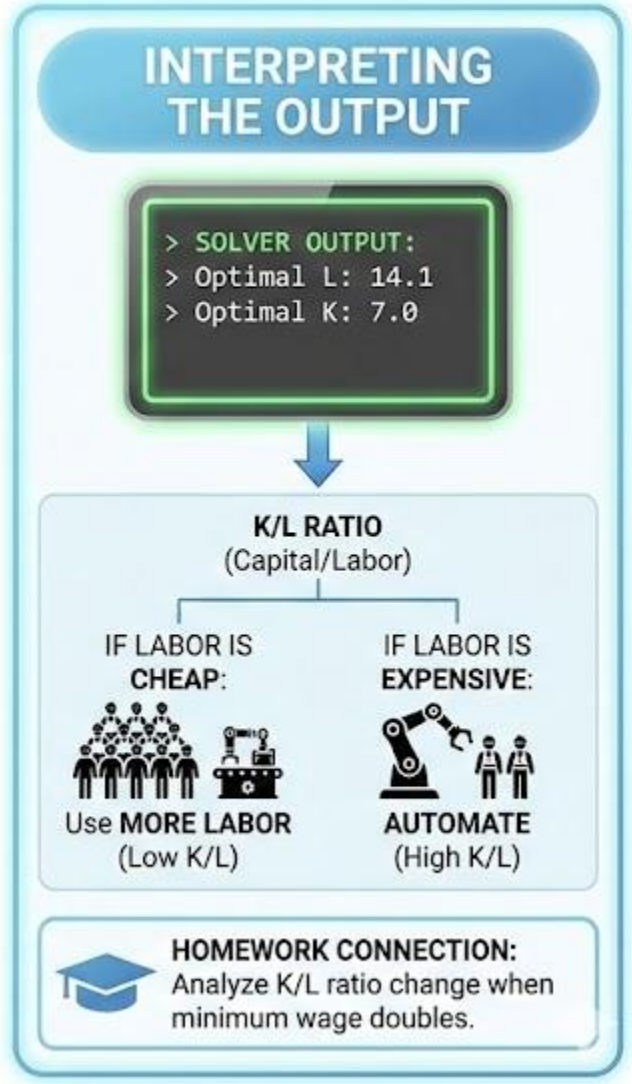
  Production Function = Target

- The solver uses numerical methods (SLSQP) to find the exact ***L*** and ***K*** values.



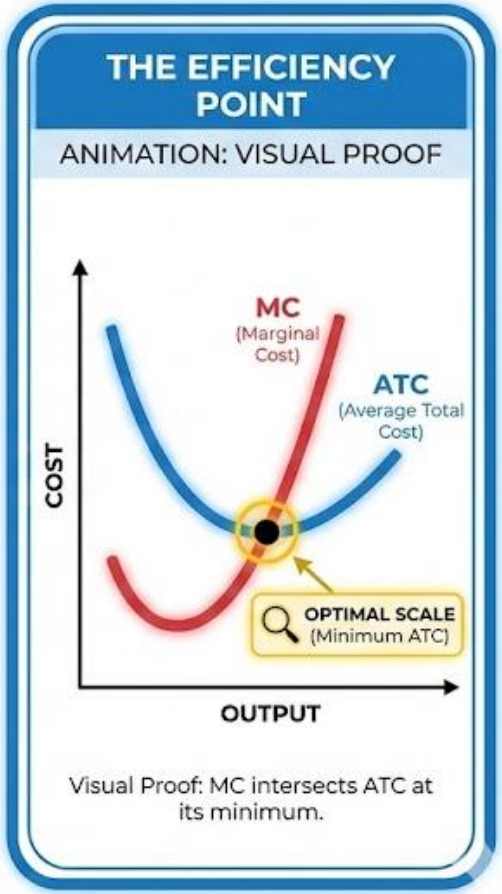
Interpreting the Output

- The solver gives us the **Optimal Input Mix**.
- **K/L Ratio:** The ratio of Capital to Labor.
  - If Labor is cheap, we use more Labor (Low K/L).
  - If Labor is expensive, we automate (High K/L).
- **Homework Connection:** You will analyze how this ratio changes when the minimum wage doubles.



Animation: The Efficiency Point

- We use matplotlib.animation to visually prove the economic theorem.
- We trace the MC and ATC curves frame-by-frame.
- **Visual Proof:** The black dot (Minimum ATC) appears exactly where the Red Line (MC) crosses the Blue Line (ATC).



Part 6: Real World & Homework

Real World Data: IPI

- **Industrial Production Index (IPI):** A monthly economic indicator measuring real output in manufacturing.
- **Dataset:** We use synthetic data mirroring the Thai OIE (Office of Industrial Economics) trends.
- **Analysis:** We can see "Supply Shocks" (e.g., COVID-19 dip) where the production capability dropped due to external constraints.



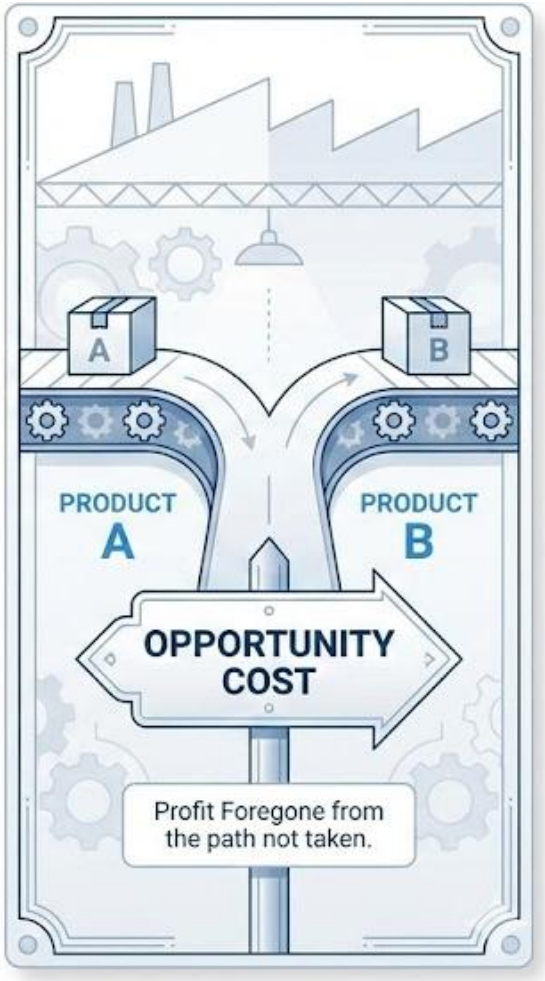
Homework Task 1: The Minimum Wage Hike

- **Scenario:** Government doubles the minimum wage.
- **Simulation:** You must run the optimizer with the new wage.
- **Critical Thinking:**
  - Does the firm just pay more?
  - Or does the firm **substitute** away from the expensive input?
  - Look at the change in the **K/L Ratio**.



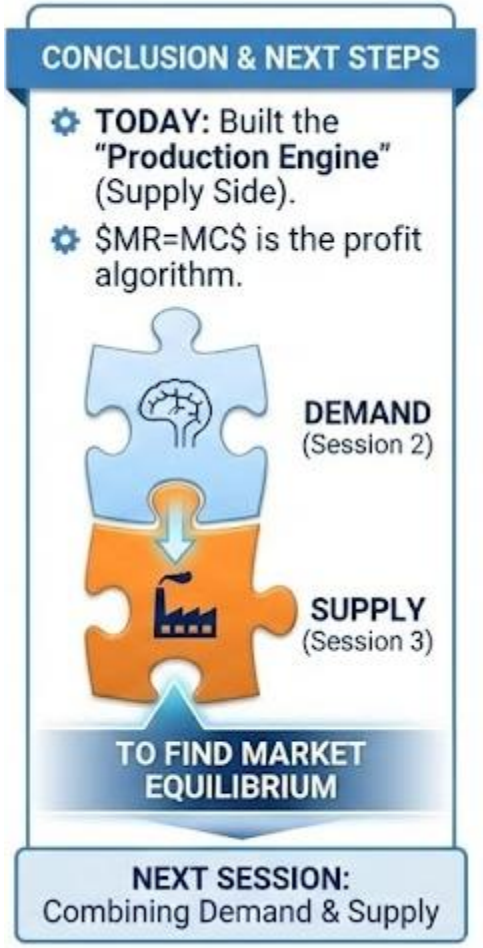
Homework Task 2: Opportunity Cost

- **Scenario:** A factory has limited floor space.
- **Choice:** Produce Product A vs. Product B.
- **Question:** Explain the Opportunity Cost.
- **Hint:** It is not about the money spent on raw materials. It is about the **Profit Foregone** from the product you *didn't* make.



Conclusion & Next Steps

- **Today:** We built the "Production Engine" (Supply Side).
- **Key Takeaway:** MR=MC is the profit algorithm.
- **Next Session:** We combine Demand (Session 2) and Supply (Session 3) to find **Market Equilibrium**.



Let’s check out our Notebook

